EXPT NO: 10	A PYTHON PROGRAM TO IMPLEMENT THE KUHN-TUCKER CONDITIONS LAGRANGIAN MULTIPLIER METHOD
DATE:	

AIM:

The objective of this lab is to introduce and implement the Kuhn-Tucker conditions for constrained optimization using Python. Participants will learn the theoretical basis of the Kuhn-Tucker conditions and apply them to solve optimization problems with inequality constraints.

PROCEDURE:

A.OBJECTIVE FUNCTION AND CONSTRAINTS:

- Define the objective function and inequality constraints in Python.

B.KUHN-TUCKER CONDITIONS:

- Implement the Kuhn-Tucker conditions using Python functions.
- Compute the gradients of the objective function and constraints.

C.OPTIMIZATION SETUP:

- Use sympy for symbolic mathematics operation or another suitable optimization function.
- Incorporate the Kuhn-Tucker conditions as additional constraints or through custom callback functions.

D.RUNNING THE OPTIMIZATION:

- Execute the optimization procedure and monitor convergence.
- Validate results against theoretical expectations.

PROBLEM -1:

Variable	Type	Condition
x_1, x_2	Real	$x_1, x_2 \geq 0$
g(x)	Linear	$x_1 + x_2 \leq 1$
f(x)	Convex	Quadratic Form

CODE

```
import sympy as sp

X1, X2 = sp.symbols('X1 X2', real=True) l1 = sp.symbols('l1', real=True)

f = X1**2 + X2**2 g1 = X1 + X2

L = f + l1 * g1

grad_L = [sp.diff(L, var) for var in [X1, X2]]

kkt_eqs = [grad_L[0], grad_L[1], l1 * g1]

solutions = sp.solve(kkt_eqs, [X1, X2, l1], dict=True)

feasible = [] for sol in solutions: g1_val = g1.subs(sol) l1_val = sol[l1] if g1_val <= 0 and l1_val >= 0:
feasible.append(sol)

if feasible: for i, sol in enumerate(feasible): print(f"\nSolution {i+1}:") x1_val = sol[X1] x2_val = sol[X2]
l1_val = sol[l1]

  print(f"Optimal point: x1={x1_val}, x2={x2_val}")
   print(f"Lagrange multiplier λ={l1_val}")

   grad_f = [sp.diff(f, var) for var in [X1, X2]]
   grad_f_val = [g.subs({X1: x1_val, X2: x2_val}) for g in grad_f]

   print(f"Gradient of f at optimal point: {grad_f_val}")

else: print("No feasible KKT solution found")
```

OUTPUT

Solution 1:

Optimal point: x1=0, x2=0

Lagrange multiplier λ=0

Gradient of f at optimal point: [0, 0]

PROBLEM 2:

Minimize:

$$f(x1,x2)=x12+x22f(x1\ ,x2\ )=x12\ +x22$$

Subject to:

$$x1+2x2\leq 2x1\ +2x2\ \leq 2$$

Solve using **KKT conditions**

CODE

```
import sympy as sp

X1, X2 = sp.symbols('X1 X2', real=True) l1 = sp.symbols('l1', real=True)

f = X1**2 + X2**2 g1 = X1 + 2*X2 - 2

L = f + l1 * g1

grad_L = [sp.diff(L, var) for var in [X1, X2]]

kkt_eqs = [grad_L[0], grad_L[1], l1 * g1]

solutions = sp.solve(kkt_eqs, [X1, X2, l1], dict=True)

feasible = [] for sol in solutions: g1_val = g1.subs(sol) l1_val = sol[l1] if g1_val <= 0 and l1_val >= 0:
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l1_val = sol[l1]

   print(f"Optimal point: x1={x1_val}, x2={x2_val}")
    print(f"Lagrange multiplier λ={l1_val}")

    grad_f = [sp.diff(f, var) for var in [X1, X2]]
    grad_f_val = [g.subs({X1: x1_val, X2: x2_val}) for g in grad_f]

    print(f"Gradient of f at optimal point: {grad_f_val}")

else: print("No feasible KKT solution found")
```

OUTPUT

Solution 1:

Optimal point: x1=0, x2=0

Lagrange multiplier λ=0

Gradient of f at optimal point: [0, 0]

RESULT

Thus the experiment Kuhn Tucker Conditions- Largrangian Multiplier Method has been successfully implemented and executed